



Module 1, Lesson 2

AI 101 FOR MECHANICAL ENGINEERS

# **Automation, robots and general intelligence**

1. Automation and Robotics



2. Intelligence Classifications



3. System Verification



4. Applied Concepts

# 1 Automation and Robotics

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## Automation Core

- Executes processes with reduced manual intervention.
- Controller executes explicit, fixed rules.
- Requires no learning component.

## Robotics Core

- Physical machines that sense and act.
- May use conventional control or AI.
- Hardware designation does not guarantee ML.

Physical Execution  
of Fixed Rules

# 2 Intelligence Classifications

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## Narrow AI

- Specialized for particular tasks.
- Operates within strictly defined domains.
- Example: Classifier identifying specified defect families.



## AGI

- Proposed broad, universal capabilities.
- Definitions and tests remain contested.
- Current assistants lack human understanding.

# 3 System Verification

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# Evaluating AI Claims



- Marketing labels cannot answer functional engineering questions.
- Analyze actual system operations over terminology.
- Example: Fluent text generation is not engineering competence.
- Chatbots cannot authorize pressure-vessel designs.

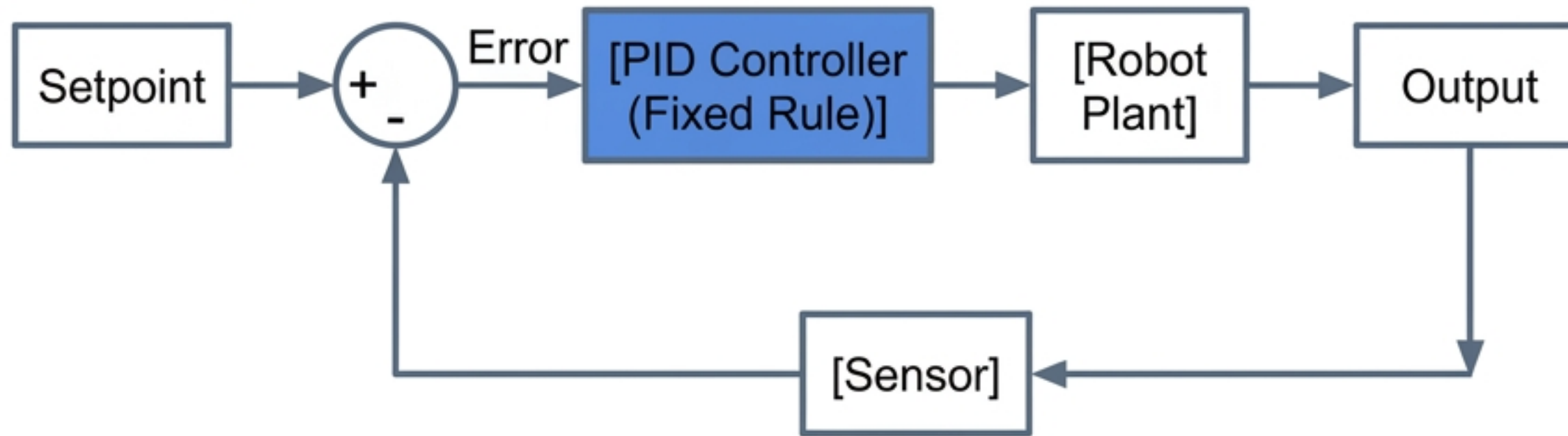
# System Verification

| System Type            | Established Function        | Required Evidence                     |
|------------------------|-----------------------------|---------------------------------------|
| Fixed Controller       | Specified automation rule   | Whether a component learns            |
| Robot Arm              | Physical sensing or action  | Its specific control mechanism        |
| Specialized Classifier | Narrow task execution       | Domain-specific performance           |
| AGI Claim              | A proposed broad capability | Agreed definitions and suitable tests |

# 4 Applied Concepts

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# System Analysis: PID Controllers



- Conventional PID robots do not necessitate ML.
- Controllers follow specified, explicit feedback rules.
- No component inherently learns from data.
- Learned components must be explicitly identified.

## Panel A: The Fixed Thermostat

Condition: Switches off above 30°C.

Diagnosis: Performs Automation.

Rationale: Explicit threshold rules establish automation, not learning.

## Panel B: The AGI Claim

Condition: Assessing AGI capabilities.

Diagnosis: Proposed broad capability.

Rationale: No single universally agreed test exists.

# Key Points Summary

1. **Automation and robotics** operate independently of **machine learning systems**.
2. **Narrow AI** executes specific tasks; **AGI** remains a contested theoretical capability.
3. Verification requires analyzing system inputs, outputs, and functional realities.
4. **Conventional feedback loops** execute fixed explicit rules without **data-driven learning**.

# Sources and Further Reading

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Primary source: AI 101 for Mechanical Engineers, Module 1, Lesson 2. Note: Worked cases are synthetic teaching material; they do not validate real equipment.

Google Machine Learning Crash Course

scikit-learn User Guide